

DECODING

LLM

By

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AI is MAGIC

(Just kidding, no it's not magic)

How many **AI** models you have used ?

May be a lot, chatGPT, claude, gemini, qwen, meta AI.

But have you ever wondered how they work internally. How is any AI model can be so good at responding to our any query.

Let's **DECODE** *it*

To understand the internal working of an LLM we first need to learn a few terms, just like to impress your crush you need to know their likes and dislikes.

Large Language Model (LLM) :

An LLM is a type of machine learning model which was trained on a huge dataset maybe in billions and trillions and is capable of answering various questions just like any human.

Tokens :

Tokens are nothing but words. For example : "Hello how are you". Here I have 4 token -> 4 words.

Vector points :

Vectors are like decimal number number points wherr each decimal point signifies a specific meaning.

Tokenization :

The process of dividing words (tokens) from a sentence.

Human Brain

Hindi, English,
French, Italian,
Bengali, Urdu,
Tamil, Gujrati,
Arabian

Languages

Hi, The,
Airplane, Thank
You, Beautiful,
Priceless, Wow,
Perfect

Words

91, 66, 1/4, 7.9,
5⁵, 7¹⁷, + -

Numbers

AI Brain

1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0
1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0
1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0
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1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0 1,0
1,0 1,0

Boolean

As human we can know about words, sentences, numbers and their operations and what they mean.

But in terms of machines, they don't know what is a word or any number or anything, they only understand 0 and 1. And represent everything in the form of 0 and 1.

Now a question

If machines and models only understand 0 and 1 then how they respond to our any query and give us exact answer just like any human ?

Answer is :

They respond back to us with a perfect human written text is just because they are trained with billions and trillions of data and they know when to use which word or where to give which answer.

Inspiration :

If you ask a kid about a car, definitely he will not be able to answer because till now the kid has not seen or heard about anything about a car.

But slowly if you teach the kid what is a car, and why we need it and what it can do then at a certain point if you ask the kid again then the kid can tell you what a car is.

This same goes for machines, if you **feed** different kinds of **data** in a model and **train** it then at a certain point the machine will become capable of answering your questions based on the data that you have provided earlier.

And that machine will be a **model** and feeding that machine will be known as **model training**.

Although this document is on the internal working of LLM and not on the training part so let's get back to the working section

Demo :

Suppose you opened ChatGPT and asked your question on "***Explain the steps in AI model training***".

You hit

Submit



Now, ChatGPT don't even know what you have just said, it doesn't know the meaning of "**explain**", "**the**" or anything.....

Now the fun part begins :

01

ChatGPT will break your query into tokens (words) and add them in an array. For example :

["Explain", "the", "steps", "in", "AI", "model", "training"]

This method is called tokenization where an LLM breaks a large query into individual tokens which helps the LLM to correctly process the data.

02

After tokenization, now with the help of some pre defined mechanisms the individual words are represented as numerical digits.

Explain > 2342

the > 1235

steps > 8791

....

....

03

Now each numerical digit is transformed into a dense vector array which will help the LLM to make decisions and understand the context of your question and respond effectively.

[0.88, 0.66, 0.121, 1.111, 0.477.....]

[0.08, 0.66, 0.2621, 1.1111, 0.777.....]

[0.81, 0.96, 0.221, 1.1921, 0.277.....]

.....

.....

04

Now, finally in this step the LLM matches the converted vectors into its existing vectors to find similarities and matches for your specific question.

Example :

cat = [0.211, 0.212, 0.234,]

dog = [1.134, 1.233, 1.765.....]

ai = [1.768, 1.999, 1.543, 1.980]

machine = [1.098, 1.677, 1.009, 1.365, ...]

05

Now based on the matching vectors the LLM combines all the vectors in one array structure and decodes them back in the numerical digits and after converting into numerical digits, it then again get back to original english words.

And you got your answer

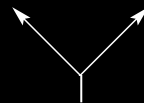
In a nutshell

Explain the steps in AI model training



["Explain", "the", "steps", "in", "AI", "model", "training"]

↓ ↓ ↓ ↓ ↓ ↓ ↓
4643 3521 9865 211 763 1146 4831



[[0.446, 0.766, 0.887, 0.311, 0.888...], [1.77,
1.0998, 1.766,]...]



*Vector Matching Algorithms &
Contextual Understanding*



[[0.486, 0.116, 0.897, 0.517, 0.248...], [1.97,
1.098, 1.756,]...]



6641 8743 46 54 246 4677 2906
↓ ↓ ↓ ↓ ↓ ↓ ↓
["The", "steps", "in", "AI", "model", "training", "involves",]

Things to keep in mind

LLMs are good for research, look for answers quickly, idea generation but you have to keep in mind that LLMs are after all machines and they can do mistakes as well.

So use LLMs effectively and don't rely too much on them. They do hallucinate, sometimes generate false out of context answers and misleading facts too.



Always remember, LLMs are just like your best friend, always confident but not always right. Verify before sharing.